

A BEGINNER'S MANUAL

Custom GPTs, Gems & Claude Projects

Build your own reusable AI assistants in ChatGPT, Gemini, Claude, Perplexity and Grok, including the 2026 move to plugins and skills

For anyone tired of pasting the same instructions into every chat • October 2026 edition • Print edition · US Letter

GUIDES FOR EVERYTHING

Custom GPTs, Gems & Claude Projects

Build your own reusable AI assistants in ChatGPT, Gemini, Claude, Perplexity and Grok, including the 2026 move to plugins and skills

Every major AI chat app lets you save a "custom assistant": a set of instructions, reference files and tools you can reuse instead of explaining yourself from scratch each time. ChatGPT called them GPTs, Gemini called them Gems, and Claude uses Projects and Skills.

2026 is a year of big change. **OpenAI is retiring custom GPTs in favor of plugins (skills plus apps). Google is turning Gems into Skills, starting with personal accounts in November 2026.** Claude Projects and Skills keep growing. This manual explains what still works, what is going away, and how to build assistants in a way that survives the next change: clear instructions, clean knowledge files and a portable `SKILL.md` format.

Every feature and date here was checked against official help-center pages on October 16, 2026, and anything we could not confirm is labeled.

HOW TO READ THIS MANUAL

- **Tip** — a shortcut or good habit.
- **Note** — useful background.
- **Warning** — easy to get wrong.
- **Danger** — can lose data, money, or access.
- `$ command` — text you type, exactly as shown.
- Settings › Privacy — a menu path to click.

New words appear in **bold** the first time and are collected in the Glossary at the back.

Features, plan availability and retirement dates were checked against official OpenAI, Google, Anthropic, Perplexity and xAI help pages on October 16, 2026. These products change monthly, and rollouts vary by account and region. Confirm details in the linked help articles before relying on them. Where official pages disagreed or no official page was found, this book says so.

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CHAPTER 1

What a Custom Assistant Really Is

- Every custom assistant, in every app, is built from the same three ingredients: **instructions**, **knowledge** (reference files) and **tools** (web search, connected apps, code).
- Different companies use different names: GPTs, plugins, Gems, Skills, Projects, Spaces.
- If you write good instructions and keep clean knowledge files, you can rebuild your assistant anywhere in minutes.

Imagine hiring a new assistant. On day one you explain who you are, how you like things written, what the job involves and where the reference documents live. A custom AI assistant is that briefing, saved once and reused forever.

1.1 The three ingredients

INGREDIENT	WHAT IT IS	EXAMPLE
Instructions	Standing orders the AI reads before every answer	"You are my meal planner. I am vegetarian. Keep recipes under 30 minutes."
Knowledge	Files the AI can look up	Your recipe collection, a style guide, a price list
Tools	Abilities beyond writing text	Web search, image creation, running code, reading your Google Drive

Every product in this book combines these three in its own way. Once you see the pattern, the differences become details.

1.2 The names, decoded

COMPANY	OLDER NAME	CURRENT OR UPCOMING NAME	WHAT IT HOLDS
OpenAI (ChatGPT)	Custom GPTs	**Plugins**, which contain **skills** and **apps**; also **Projects**	Instructions, files, connected apps
Google (Gemini)	Gems	**Skills**	Instructions, plain-text files
Anthropic (Claude)	n/a	**Projects** and **Skills**	Instructions, knowledge, packaged workflows
Perplexity	Spaces	**Projects**	Instructions, files, links
xAI (Grok)	n/a	**Projects** (grok.com)	Instructions and files (details unconfirmed)

1.3 Two shapes of assistant

It helps to sort assistants into two shapes:

1. **Workspace assistants** (Projects). A folder that holds instructions, files and a group of related chats. Everything you do inside the folder uses its setup. Good for ongoing work: a house renovation, a class, a client.
2. **Skill assistants** (Skills, plugins, Gems, GPTs). A packaged "how to do this task" that you can call up from any chat. Good for repeatable jobs: "turn notes into a weekly report," "check this email for tone."

Many people use both: a Project for each big area of life, and a handful of Skills for tasks that come up everywhere.

> **TIP**
> If you find yourself pasting the same paragraph of instructions more than three times a week, that paragraph should become a Project instruction or a Skill.

1.4 What custom assistants cannot do

- **They do not train the model.** Your instructions and files are read at the time of the chat. The underlying AI is not retrained on them.
- **They do not guarantee accuracy.** An assistant with a perfect price list can still misread it. Check important answers.
- **They do not remember everything forever.** Files and instructions persist, but individual chats may not share memory with each other unless the product says so. Claude's help center, for example, notes that context is not shared across chats in a project unless it is added to the project knowledge.
- **They are not private vaults.** Files you upload are stored by the provider under its privacy terms. Do not upload what you would not want stored in that service.

- [ ] I can name the three ingredients: instructions, knowledge and tools.
- [ ] I know the difference between a workspace assistant (Project) and a skill assistant.
- [ ] I know custom assistants do not retrain the AI.

CHAPTER 2

The Big 2026 Changes, Explained

- **ChatGPT:** custom GPTs retire on **December 11, 2026** (February 11, 2027 for Enterprise workspaces with an approved deferral). The replacement is **plugins**. Personal accounts (Free, Go, Plus, Pro) can no longer create new GPTs.
- **Gemini:** Gems transition to **Skills** for personal Google accounts starting **November 2026**, and for business and school accounts in 2027. Google says it will recreate your Gems as skills automatically.
- **Claude:** Projects are available on every plan (Free is limited to five), and Skills are available on all plans with code execution turned on. A new version of Projects is in beta.

If you read an older article about "building a GPT" or "making a Gem," parts of it are now out of date. Here is the state of play on October 16, 2026.

2.1 ChatGPT: from GPTs to plugins

OpenAI's help center states that custom GPTs are being retired and that creators should migrate to **plugins**. Key points from the official *Custom GPT retirement and migration FAQ* and *GPTs in ChatGPT* articles:

- **Retirement date:** December 11, 2026. Enterprise workspaces with an approved deferral get until February 11, 2027.
- **Until then:** existing GPTs keep working, and you can keep editing a GPT you have not migrated.
- **At retirement:** custom GPTs stop running and leave the GPT directory.
- **Migration:** creators go to **My GPTs** and choose **Migrate to plugin** for a published GPT, when the option reaches their account. OpenAI targeted the migration experience for September 17, 2026, rolling out gradually.

What moves across and what does not:

GPT PART	AFTER MIGRATION
Instructions	Become a **skill** inside the plugin
Knowledge files	Become reference files
Connected apps	Added to the plugin as apps
Custom actions (your own API connections)	**Not migrated**; rebuild separately
Conversation starters	Not carried over one-to-one
Selected model	Not carried over
Past chats and sharing settings	Not carried over

> ☆ IMPORTANT
>
> After you migrate a GPT, OpenAI says the original becomes **read-only**. Test the new plugin before you rely on it, and save copies of your instructions and conversation starters somewhere safe first.

> ⚠ WARNING
>
> **Conflicting official information (checked 2026-10-16):** OpenAI's help center says personal accounts (Free, Go, Plus and Pro) can no longer create new GPTs, but the public pricing page still listed custom GPT creation as a Plus feature on the same day. Treat the help center, which describes the retirement in detail, as the more current source, and check your own account.

2.2 Gemini: from Gems to Skills

Google's *About the transition from Gems to skills* article says:

- **Personal Google accounts:** Gems transition to skills starting **November 2026**.
- **Work and school accounts:** in 2027. Google's Workspace admin article gives "no sooner than March 1, 2027" for business and enterprise, and lists education later.
- **Automatic migration:** when Gems are removed, Google recreates them as skills. The Workspace article says migrated Gems arrive as **inactive drafts** that the creator must review and turn on.
- **Manual migration:** you can recreate a Gem as a skill yourself at any time (steps in the Gemini chapter).
- **What may not carry over:** Google notes GitHub files are not supported in skills, and only certain file types can be uploaded.

2.3 Claude: Projects and Skills grow up

Anthropic did not retire anything. Instead:

- **Projects** are available to all users. Free users can create up to five.
- **Retrieval (RAG) for projects** is available on paid plans and switches on automatically when project knowledge gets large, expanding capacity "by up to 10x."
- **Skills** are available on Free, Pro, Max, Team and Enterprise, as long as **Code execution and file creation** is enabled.
- A **new version of Projects** is in beta, rolling out first to Claude Code users and some Pro and Max accounts.

2.4 What this means for you

1. **Write your instructions in a plain text file** you own, not only inside an app. You will reuse them.
2. **Keep your knowledge files in a folder** on your computer or cloud drive, with clear names.
3. **Prefer formats that travel.**
   Markdown (`.md`) and plain text work almost everywhere; Gemini skills accept only plain-text-based files.
4. **Learn the `SKILL.md` format.**
   Claude Skills, Gemini Skills and Grok Build all use a folder with a `SKILL.md` file, so one well-written skill can move between them with small edits.

- [ ] I know custom GPTs retire on December 11, 2026 for most accounts.
- [ ] I know personal Gems become skills from November 2026.
- [ ] I know Claude Projects work on the Free plan (up to five).
- [ ] I keep my own copy of every instruction set I write.

CHAPTER 3

Writing Instructions That Work

- Good instructions answer five questions: **who** the assistant is, **who** it is helping, **what** the job is, **how** to format answers, and **what not** to do.
- Short, specific rules beat long, vague essays.
- Include one example of a perfect answer.

The quality of a custom assistant is mostly the quality of its instructions. The same rules work in every product.

3.1 The five-part instruction template

PROMPT

```
ROLE
You are a friendly, precise assistant who helps me plan weekly meals.

ABOUT ME
I cook for a family of four. One child is allergic to peanuts.
We like Mexican and Italian food. Budget-conscious.

THE JOB
When I ask for a plan, produce 5 dinners for Monday to Friday.
Each dinner must take 30 minutes or less.
Reuse ingredients across the week to cut waste.

FORMAT
- A table: Day | Dish | Time | Key ingredients
- Then a grocery list grouped by store section.
- Keep the whole answer under 350 words.

RULES
- Never include peanuts or peanut oil. Flag any "may contain" risk.
- If I give leftovers or what is in my fridge, use those first.
- If something is unclear, ask one short question before planning.
```

3.2 Why each part matters

PART	PURPOSE	COMMON MISTAKE
Role	Sets tone and expertise	“You are the world’s greatest genius” adds nothing
About me	Gives context the AI cannot guess	Leaving out constraints such as allergies
The job	Defines what success looks like	Describing a vibe instead of a task
Format	Makes answers consistent	No format, so every answer looks different
Rules	Prevents known problems	Too many rules, some contradicting others

3.3 Ten rules for strong instructions

1. **Be specific.** “Under 200 words” beats “be concise.”
2. **Use plain lists.** Bullet points are easier for the AI to follow than long paragraphs.

3. **Put the most important rule first** and repeat critical safety rules at the end.
4. **Give one example** of an ideal answer.
5. **Say what to do, not only what to avoid.** “Write at an 8th-grade reading level” is clearer than “don’t be complicated.”
6. **Tell it when to ask questions** instead of guessing.
7. **Tell it how to use the files.** “Answer from the price list; if the item is not listed, say so.”
8. **Avoid contradictions.** “Be brief” plus “explain everything thoroughly” confuses it.
9. **Name the audience.** “My readers are retirees new to smartphones.”
10. **Version your instructions.** Keep a dated copy each time you change them.

3.4 Instructions for using knowledge files

The single most valuable instruction for any assistant with files:

PROMPT

```
Use the attached files as your primary source. When you use them,
name the file you used. If the answer is not in the files, say
"That isn't in my files" and then, only if I ask, answer from general
knowledge, clearly labeled as such. Never invent prices, dates or policies.
```

3.5 Instructions that reduce mistakes

PROMPT

```
Before answering anything involving numbers, show the calculation.
If you are less than confident, say so and explain what would confirm it.
Do not make up quotes, statistics, links or sources.
```

> **TIP**
>
> Treat your first version as a draft. Use the assistant for a week, note every answer you had to correct, and add one rule per problem. Most good assistants reach their final form after three or four revisions.

3.6 Description vs. instructions

Several products now ask for a short **description** as well as instructions. They serve different purposes:

- The **description** helps the app (or the AI) decide **when** to use the assistant. Gemini's help center says it checks only a skill's name and description to decide whether the skill is relevant. Claude's troubleshooting advice is similar: if Claude is not using a skill, check that the description clearly explains when to use it.
- The **instructions** tell the AI **how** to do the job once it is chosen.

A good description is one or two sentences naming the trigger: "Use when the user asks to turn meeting notes into an action-item summary for the weekly team email."

- [ ] My instructions cover role, audience, job, format and rules.
- [ ] I included one example of an ideal answer.
- [ ] I told the assistant how to use its files and what to do when the answer is not there.
- [ ] My skill descriptions name the situations that should trigger them.

CHAPTER 4

Preparing Knowledge Files

- The AI can only use what it can read. Clean, well-labeled text beats messy scans.
- Plain text and Markdown work almost everywhere; check each product's file rules before you upload.
- Split large collections into focused files and name them clearly.

4.1 What makes a good knowledge file

GOOD	BAD
Clear headings and short sections	One giant wall of text
Real text you can select and copy	A scanned image of a page
One topic per file	A random mix of unrelated material
A date and version at the top	No way to tell if it is current
File names like `returns-policy-2026-10.md`	`Document (3) final FINAL.pdf`

4.2 Converting documents to clean text

- **Word documents:** most products accept `.docx`, but saving as `.md` or `.txt` makes them portable.
- **PDFs:** many products accept PDFs, but scanned PDFs (images of text) may read poorly. If you cannot select the text in a PDF viewer, run it through a text-recognition (OCR) tool first.
- **Spreadsheets:** export a clean `.csv` with one header row. Remove merged cells and decorative formatting.
- **Web pages:** copy the useful text into a `.md` file and note the source URL and date at the top.

4.3 Product file rules (checked 2026-10-16)

PRODUCT	WHAT THE HELP CENTER SAYS
Gemini Skills	Plain-text-based files only (for example `.txt`, `.md`, `.csv`, `.json`, `.yaml`, `.py`, `.html`); total upload up to 100 MB; scripts that need the internet are not supported
Claude Projects	Upload documents, text files or code; paid plans switch to RAG automatically when knowledge nears the context limit
Perplexity Projects	Up to 10,000 files and 5 GB per Project, 100 MB per file, the same on every plan; no restriction on file type
ChatGPT	File types and limits vary by feature; see OpenAI's file upload FAQ

4.4 A knowledge file template

RETURNS-POLICY-2026-10.MD

```
# Returns policy (version 2026-10, approved by J. Smith)

## Summary
- Returns accepted within 30 days with receipt.
- Store credit only after 30 days, up to 60 days.

## Exceptions
- Opened software: exchange only.
- Clearance items: final sale.

## How to process a return
1. Check the receipt date.
2. Inspect the item.
3. ...
```

> ⚠ **WARNING**
>
> Remove private information you do not need before uploading: account numbers, health details, other people's contact information, passwords. If a file is shared along with an assistant, anyone who can use the assistant may be able to get the AI to quote it.

- [ ] My files are real text, not scanned images.
- [ ] Each file covers one topic and has a date at the top.
- [ ] I removed unnecessary private information.
- [ ] I checked the product's file limits before uploading.

CHAPTER 5

ChatGPT: Projects, GPTs and Plugins

- **Projects** are the everyday way to build a reusable workspace in ChatGPT: a folder with its own instructions, files and chats. Projects are available on the Free plan.
- **Custom instructions** (in personalization settings) apply to all your chats.
- **GPTs** still run until December 11, 2026, but personal accounts cannot create new ones. Migrate the ones you care about to **plugins**.
- **Plugins** bundle skills and apps. You create them by chatting with **@plugin-creator**.

5.1 Option 1: ChatGPT Projects (best for most people)

A Project groups related chats with shared files and instructions. OpenAI's plan comparison lists Projects on the Free plan, so anyone can use them.

1. In the ChatGPT sidebar, choose to create a **new project** and give it a name, for example "Kitchen Renovation."
2. Open the project's settings and add **instructions**. Paste your five-part instructions from the previous chapter.
3. Add **files**: quotes from contractors, measurements, a budget spreadsheet.
4. Start chats inside the project. Each one sees the project's instructions and files.

Good uses for ChatGPT Projects:

- A class, with the syllabus and lecture notes as files.
- A job search, with your résumé and target job descriptions.
- A household project, with quotes, receipts and to-do lists.
- A writing project, with a style guide and chapter outlines.

> **NOTE**
> ChatGPT's file limits, project features and memory options differ by plan and change often. If an option described here is missing, check OpenAI's help center article on Projects for your plan.

5.2 Option 2: Custom instructions for every chat

Custom instructions live in ChatGPT's personalization settings and apply to every new chat outside of special contexts. Use them for things that are always true:

PROMPT

```
About me: I'm a high-school science teacher in Indiana. I prefer
US spelling and units, short paragraphs and bullet points.
How to respond: Be direct. When you're unsure, say so. Don't use
emojis. Offer one follow-up suggestion at the end.
```

Keep task-specific rules out of custom instructions. They belong in a Project or a plugin skill, or they will leak into unrelated chats.

5.3 Option 3: Existing GPTs (until retirement)

If you already built GPTs, they keep working until the retirement date for your account. A GPT's parts, as described in OpenAI's *GPTs in ChatGPT* article:

PART	WHAT IT DOES
Instructions	Behavior, tone, goals and boundaries
Conversation starters	Example prompts shown to users
Knowledge	Uploaded reference files
Capabilities	Tools such as web search or image generation
Apps	Connections to outside services
Actions	Connections to external APIs that the creator defines

> **DANGER**
> OpenAI warns that when a GPT uses apps or external APIs, relevant parts of your input may be sent to that third-party service, and OpenAI does not audit or control how those services use your data. Only use GPTs with actions from creators you trust.

5.4 Migrating a GPT to a plugin

When the migration option reaches your account:

1. **Back up first.**
   Copy the GPT's instructions, description and conversation starters into a text file. Download its knowledge files.
2. Go to **My GPTs** and choose **Migrate to plugin** next to a published GPT. (Drafts cannot be migrated directly; OpenAI notes that publishing does not require sharing publicly.)
3. Review the migration details and follow the steps.
4. **Rebuild custom actions** separately. They do not migrate.
5. **Recreate conversation starters** as saved prompts or as examples inside the skill instructions.
6. **Test** the plugin on your usual tasks. The model is not carried over, so answers may differ.
7. **Share** the plugin if you are in a Business or Enterprise workspace and have permission.

Only the GPT's creator, or a workspace owner or admin, can migrate it. Having access to someone else's GPT is not enough. If you rely on someone else's GPT, watch for the creator's notice about a replacement.

5.5 Plugins: the new building block

OpenAI's *Plugins in ChatGPT* article defines a plugin as a reusable package that can include:

- **Skills:** instructions and workflow guidance (the part that replaces GPT instructions).
- **Connected apps:** links to external accounts, information and actions, such as Slack or Google Drive.
- **App templates:** for administrators configuring apps in a workspace.
- **Extensions:** interactive views, file editing, forms and settings.

Finding and using plugins

- Open **Plugins** in the ChatGPT sidebar (web, desktop or mobile) to browse the directory.
- Install a plugin, and connect any apps it needs.
- ChatGPT **uses installed plugins automatically** when relevant. You can also call one with an **@ mention**, by selecting **+** and typing its name, or with **Plugins** in the message box.
- Some plugins are marked **Desktop only** and need the ChatGPT desktop app.

Creating a plugin with Plugin Creator

1. Install **Plugin Creator** from the plugin directory.
2. Start a chat with **@plugin-creator** and describe what you want, including any template or example output.
3. Answer its questions and refine the instructions.
4. Install the result if prompted, and connect any apps.
5. Test it by mentioning it with @ and checking the output.
6. To change it later, mention **@plugin-creator** and the plugin name, and describe the edit.

OpenAI's example request:

PROMPT

Create a Project Updates plugin that gathers progress from Slack and Linear and drafts a weekly update with key milestones, blockers, and next steps. Add a skill that uses the attached template to keep updates consistent.

IMPORTANT

Plan availability is uneven (checked 2026-10-16). The plugin directory is available across ChatGPT plans, but OpenAI says installing or using a particular plugin depends on your plan, workspace, role and region, and that **personal Skills are generally available on Business, Enterprise, Healthcare and Edu**. On Free, Go, Plus or Pro, check the Plugins area of your own account to see what you can create. For most personal-plan users today, **Projects** are the dependable way to save instructions and files.

5.6 Choosing the right ChatGPT tool

YOU WANT TO...	USE
Set your general writing preferences once	Custom instructions
Work on an ongoing topic with files	A Project
Keep using a GPT you already built	The GPT, until retirement; then migrate
Package a repeatable workflow, possibly with apps	A plugin (if available on your plan)
Share a workflow with your team	A plugin shared in a Business or Enterprise workspace

- [ ] I created a ChatGPT Project with instructions and files.
- [ ] I set custom instructions for things that are always true about me.
- [ ] I backed up every GPT I built (instructions, starters, files).
- [ ] I know how to find Plugins and @plugin-creator.

CHAPTER 6

Gemini: Gems and the Move to Skills

- **Skills** are Gemini's reusable instructions. Create them at **Settings > Skills** on gemini.google.com, in the Gemini app on Mac, or in the mobile app.
- A skill is a folder with a `SKILL.md` file and optional plain-text reference files. Names use lowercase words joined by hyphens.
- **Gems** still exist for now, but personal-account Gems transition to skills from November 2026. Google says it will recreate them automatically.

6.1 How Gemini skills work

Google's *Write effective skills* article explains that Gemini reads skills in stages:

1. It checks only the skill's **name and description** to decide whether the skill is relevant.
2. If it is, Gemini reads the **full instructions**.
3. It opens any **attached files** (a template, a style guide, a spreadsheet) only when needed.

You can also call a skill directly by typing **/** in the chat box. Google's help center says skills on personal accounts are for users 18 and older. The availability list in Google's help center says skills are in the Gemini mobile app, the Gemini app on Mac and the web app at gemini.google.com.

6.2 Ways to create a skill

From the **Skills** page (Settings > Skills), Google lists several options:

- **Create with help from Gemini:** describe what you want, and Gemini drafts the skill.
- **Start from a template:** prefilled or blank.
- **Create manually:** type a name, description and instructions.

- **Upload:** a `SKILL.md` file, or a folder or `.zip` with `SKILL.md` in its main folder.
- **From a chat or task:** ask Gemini to turn what you just did into a skill.

6.3 Create a skill manually

1. Go to gemini.google.com and sign in.
2. In the sidebar, open **Settings**, then **Skills**.
3. Click **Create manually**.
4. Enter a **name** (Gemini converts it to the lowercase-hyphen format), a one- or two-sentence **description** and your **instructions**.
5. Click **Create**.
6. Test it in a new chat by describing a matching task, or by calling the skill directly.

6.4 The `SKILL.md` file

When you download or upload a skill, it is a folder whose main file is `SKILL.md`. A simple example that works as a starting point for Gemini, Claude and Grok Build:

WEEKLY-REPORT/SKILL.MD

```
---
name: weekly-report
description: Use when the user asks to turn rough notes, chat logs or bullet points into the Friday
team update email.
---

# Weekly report

## Steps
1. Read the notes and group them into Done, In progress and Blocked.
2. Write a 2-sentence summary at the top.
3. Use the format in template.md exactly.
4. Keep it under 250 words.

## Rules
- Do not invent progress that is not in the notes.
- If a blocker has no owner, add "(owner needed)".
```

Google's rules for uploaded skills (checked 2026-10-16):

RULE	DETAIL
Required file	`SKILL.md` in the skill's root (main) folder

Name format	All lowercase, words separated by hyphens, such as `my-new-skill`
Folder name	Must match the skill name when you upload a folder
Total size	All uploaded files together must not exceed 100 MB
File types	Plain-text-based files: text and docs (`.txt` , `.md` , `.rtf`), code (`.py` , `.sh`), data (`.json` , `.yaml` , `.csv` , `.xml` , `.sql`), web (`.html` , `.css` , `.svg`)
Scripts	Scripts that need internet access are not supported
Uploading	Currently on the web app and the Mac app

> **TIP**
> Google's rule of thumb: if you can open a file in a basic text editor and read it, it is probably supported. That means PDFs and Word files need converting to `.md` or `.txt` first.

6.5 Gems (legacy)

Gems are the older format: a name, instructions and optional knowledge files, created in Gemini's Gem manager. They still work for now, but:

- **Personal accounts:** Gems transition to skills from **November 2026**.
- **Business and enterprise Workspace accounts:** no sooner than **March 1, 2027**, according to Google's Workspace admin article. After that date, users can no longer create, edit or use Gems, and Workspace Studio flows with an "Ask a Gem" step stop working.
- **Education accounts:** later in 2027.

6.6 Recreate a Gem as a skill yourself

Google's steps, summarized:

1. Open the Gem and copy its **name**, **description** and **instructions** into a text file. Download each knowledge file to one folder on your computer.
2. In a new tab, go to gemini.google.com, then **Settings > Skills > Create manually**.
3. Paste the name, description and instructions, then click **Create**.
4. On the Skills page, hover over the new skill, open **Skill actions** and choose **Download**. This saves a `.zip` containing `SKILL.md`.
5. Unzip it and move `SKILL.md` into the folder with your Gem's files. **Rename the folder to match the skill name.**

6. Back on the Skills page, choose **Skill actions > Replace skill**, select **folder**, and upload the folder.

> **NOTE**
> If you do nothing, Google says it will recreate your Gems as skills automatically. For Workspace accounts, Google says migrated Gems arrive as inactive drafts that you must review and activate. Doing it yourself lets you clean up the instructions and convert files to supported formats in advance.

6.7 Writing skills Gemini actually uses

Google's best-practice advice, in plain terms:

- Make the **name and description specific**. A vague description means Gemini may not pick the skill when you need it.
- Write instructions for a **type of task**, not one single request.
- Include **clear examples**, especially for formatting.
- For long or multi-step jobs, give **numbered steps**.

- [ ] I found Settings > Skills in Gemini.
- [ ] I created one skill manually and tested it.
- [ ] I backed up my Gems' instructions and files.
- [ ] My skill files are plain text and under 100 MB in total.

CHAPTER 7

Claude: Projects and Skills

- **Projects** are self-contained workspaces with their own chats, knowledge and instructions. Everyone can use them; Free accounts can create up to five.
- Paid plans switch project knowledge to **RAG** automatically when it grows large, expanding capacity by up to 10x.
- **Skills** package instructions, scripts and resources. Enable **Code execution and file creation** in **Settings > Capabilities**, then manage skills in **Customize > Skills**.

7.1 Create a Claude Project

From Anthropic's *How can I create and manage projects?* article:

1. Click **Projects** in the left sidebar, or go to claude.ai/projects.
2. Click **+ New Project**.
3. Enter a name and description. (Anthropic notes that Claude does not see the name and description; they are labels for you.)
4. On Team or Enterprise plans, choose who can see the project.
5. Use the + button in the project knowledge area to upload documents, text or code.
6. Click **Set project instructions**, write your instructions and click **Save instructions**.
7. Start chats inside the project.

> ☆ IMPORTANT
>
> Anthropic's help center says context is **not shared across chats** within a project unless it is in the project knowledge. If a chat produces something you want future chats to know (a decision, a summary), add it to the project knowledge.

7.2 Project knowledge and RAG

- Everything in project knowledge is available to every chat in that project.
- On **Free**, project knowledge is limited by the model's context window.
- On **paid plans** (Pro, Max, Team, Enterprise), Claude automatically enables **RAG mode** when knowledge approaches the context limit. Instead of reading everything at once, Claude searches the knowledge and pulls in the relevant parts. Anthropic says this expands capacity by up to 10x, and you will see an indicator when a project is RAG-enabled.

> ⚠ **WARNING**
>
> **Conflicting official information (checked 2026-10-16):** Anthropic's project articles say projects, including project instructions, are available to all users, while its personalization article describes project instructions as "paid plans only." If you are on Free and do not see **Set project instructions**, put your instructions in a knowledge file named `instructions.md` instead.

7.3 Project sharing

Claude's pricing table (checked 2026-10-16) lists **project sharing and collaboration** as not included on Free, Pro or Max individual plans. Sharing is a team-plan feature.

7.4 Claude Skills

Anthropic's *Use skills in Claude* article (checked 2026-10-16):

- Skills are available on **Free, Pro, Max, Team and Enterprise**.
- They require **code execution** to be enabled.
- Anthropic provides built-in skills for **Excel spreadsheets, Word documents, PowerPoint presentations and PDFs**. Claude uses them automatically when relevant.
- You can **upload your own** skills as ZIP files.
- Skills you enable also load in **Claude Code** (version 2.1.273 or later, signed in with the same account) and in Claude's Microsoft 365 add-ins, where you can type **/** to pick a skill.

Turn skills on

1. Go to **Settings > Capabilities** and make sure **Code execution and file creation** is on.
2. Go to **Customize > Skills**.
3. Toggle the example or built-in skills you want.

On Team and Enterprise plans, an owner must enable code execution and Skills in **Organization settings > Plugins & skills** (Policy tab) first.

Upload a custom skill

1. Create a folder named exactly like your skill, for example `weekly-report`.
2. Put a `SKILL.md` inside it with `name` and `description` at the top (the same format shown in the Gemini chapter).
3. Add any templates or reference files.
4. Zip the folder.
5. In **Customize > Skills**, click **+**, then **+ Create skill**, then **Upload a skill**, and choose the ZIP.
6. Toggle it on and test it with a few different prompts.

Common upload errors, from Anthropic's troubleshooting list:

ERROR CAUSE	FIX
ZIP too large	Remove big files; link to them instead
Folder name does not match the skill name	Rename the folder to the `name` in `SKILL.md`
Missing skill file	Make sure `SKILL.md` is inside the top folder
Invalid characters in name or description	Use lowercase letters, numbers and hyphens in the name

Sharing skills

On **Team and Enterprise** plans you can share a skill with specific colleagues (view-only for them) or publish it to your organization's library, if an owner has enabled those options. On individual plans, each person uploads skills to their own account.

> **DANGER**
> Anthropic lists **prompt injection** and **data exfiltration** as the main risks of skills, and advises installing skills only from trusted sources. Before enabling a skill from anyone else, read its files, especially scripts and any instructions to connect to outside websites.

7.5 Projects or Skills?

SITUATION	CHOOSE
Ongoing work on one topic with many documents	Project
A repeatable task you do in many different contexts	Skill
A task that produces a file (spreadsheet, slides) in a set style	Skill with a template file
Sharing a workspace with teammates	Project on a Team plan
Using the same workflow in Claude Code	Skill (syncs to Claude Code)

- [ ] I created a Claude Project with knowledge files and instructions.
- [ ] I turned on Code execution and file creation.
- [ ] I uploaded one custom skill as a ZIP and tested it.
- [ ] I only enable skills whose files I have read.

CHAPTER 8

Perplexity and Grok

- **Perplexity Projects** (formerly Spaces) hold instructions of up to 8,000 characters plus files and links. File limits are the same on every plan.
- **Grok Projects** exist on grok.com, according to xAI's FAQ, but we found no official page describing their limits. Details here are labeled unconfirmed.
- **Grok Build** (xAI's coding agent) reads `AGENTS.md` rule files and `SKILL.md` skills, and is compatible with Claude Code skills and plugins.

8.1 Perplexity Projects

Perplexity's *What are Projects?* article (checked 2026-10-16) says:

- After creating a Project, you add **instructions of up to 8,000 characters** in its settings.
- Each Project can store **up to 10,000 files and 5 GB**, with each file up to **100 MB**. The article says these limits are **the same on every plan**, with no restriction on file type.
- Project settings also let you add **web links and domains to prioritize**, choose a default mode, and pick which connectors the Project uses.
- Files imported from cloud storage are **static copies**; import again to bring in a newer version.

Perplexity Projects shine for research assistants: put your reference papers in the Project, add instructions about citation style, and prioritize trusted domains.

PROMPT

```
You are my research assistant for a report on home heat pumps.
Prefer sources from energy.gov, government agencies and peer-reviewed
journals. Always cite. Separate "established facts" from "estimates."
Use US units. End with a short list of open questions.
```

8.2 Grok Projects

xAI's official Grok FAQ tells users to use **grok.com** in a standard Chrome or Chromium browser, noting that some users on other hosts "run into missing features like Projects." That confirms Projects exist on grok.com.

> **NOTE**
>
> **Unverified (2026-10-16):** We could not find an official xAI page describing Grok Projects' file limits, instruction length or plan availability. Third-party guides describe a familiar pattern: create a Project, add instructions, upload files, and start chats inside it so they inherit the setup. Use Grok Projects that way, but confirm limits in the app.

8.3 Grok Build: rules and skills for coding

xAI's developer documentation for **Grok Build** (its coding agent) describes two customization layers that use the same portable formats as other tools:

- **Project rules:** Markdown files loaded into every session in a folder tree. Grok reads `AGENTS.md` (and compatible names such as `CLAUDE.md`), plus files in `.grok/rules/`. Running `grok inspect` lists the rules files it found.
- **Skills:** folders with a `SKILL.md` file, discovered from `./.grok/skills/`, `~/.grok/skills/`, `~/.agents/skills/` and enabled plugins.
- **Compatibility:** the docs say Grok automatically reads Claude Code marketplaces, plugins, skills and instruction files alongside its own.

That means a `SKILL.md` folder you write for Claude can often be dropped into `~/.grok/skills/` unchanged.

CHAPTER 9

One Assistant, Every App: The Portable Method

- Keep a **master folder** for each assistant: instructions, a `SKILL.md`, knowledge files and test prompts.
- From that folder you can build a ChatGPT Project, a Claude Project or Skill, a Gemini Skill, a Perplexity Project and a Grok Build skill in minutes.
- When a product changes (as GPTs and Gems did), you lose nothing.

9.1 The master folder

FOLDER LAYOUT

```
meal-planner/
  SKILL.md            name, description, instructions
  instructions.txt    the same instructions without the header, for Projects
  knowledge/
    family-preferences.md
    pantry-staples.csv
    favorite-recipes.md
  tests/
    test-prompts.md   5 prompts with notes on what a good answer looks like
  CHANGELOG.md        dated notes on each change
```

9.2 Building each version from the folder

TARGET	WHAT TO COPY	NOTES
ChatGPT Project	`instructions.txt` into project instructions; `knowledge/` files into project files	Works on Free
ChatGPT plugin	Describe the task to @plugin-creator and attach `SKILL.md` and templates	Plan-dependent
Claude Project	`instructions.txt` into project instructions; `knowledge/` into project knowledge	Free: up to 5 projects
Claude Skill	Zip the folder (name must match) and upload in Customize > Skills	Needs code execution
Gemini Skill	Upload the folder in Settings > Skills	Plain-text files only; 100 MB total
Perplexity Project	`instructions.txt` (under 8,000 characters) plus files	Add priority domains
Grok Build	Copy the folder into `~/.grok/skills/`	Reads `SKILL.md`

> **TIP**
>
> Keep `instructions.txt` under 8,000 characters and you will fit Perplexity's limit, and you will also have instructions short enough that models follow them reliably.

9.3 A portable `SKILL.md` checklist

- [ ] Folder name equals the `name` field.
- [ ] `name` is lowercase with hyphens, such as `meal-planner`.
- [ ] `description` is one or two sentences that say **when** to use the skill.
- [ ] The body has numbered steps, a format section and a rules section.
- [ ] Reference files are plain text (`.md`, `.csv`, `.txt`).
- [ ] No scripts that need internet access (Gemini does not support them).
- [ ] A `tests/` file lists prompts to check after each change.

CHAPTER 10

Twelve Ready-to-Use Assistant Recipes

Each recipe gives a name, a description for skill systems, and instructions you can paste into any Project or skill. Adapt the details in brackets.

Recipe 1: Plain-English explainer

Description: Use when the user pastes a confusing document (a bill, a contract clause, a letter) and wants it explained simply.

PROMPT

```
Explain the pasted text in plain English at an 8th-grade reading level.
Structure: 1) What it is, in one sentence. 2) What it means for me,
as 3-5 bullets. 3) Anything I need to do, with deadlines. 4) Questions
I should ask. Quote the exact sentence for any deadline or amount.
Do not give legal or financial advice; suggest who to ask if it matters.
```

Recipe 2: Email tone checker

Description: Use when the user asks whether an email sounds right, or wants it rewritten to be clearer or kinder.

PROMPT

```
First, rate the draft's tone in one line (for example "polite but vague").
Then list up to 3 specific problems. Then give a rewritten version that
keeps my meaning, is 20% shorter, and ends with a clear request.
Keep my voice: casual, friendly, no corporate jargon.
```

Recipe 3: Meeting notes to action items

Description: Use when the user pastes meeting notes or a transcript and wants decisions and tasks.

PROMPT

```
Produce: Decisions (bullets), Action items (table: Task | Owner | Due),
Open questions (bullets). Only include owners and dates that appear in
the notes; otherwise write "unassigned" or "no date". Under 250 words.
```

Recipe 4: Study coach

Description: Use when the user wants to study a topic from their class files using quizzes and explanations.

PROMPT

```
Use the course files as the source of truth. Teach with the Socratic method:
ask one question at a time, wait for my answer, then explain what I got right
and wrong. Every 5 questions, give a short summary of my weak spots.
Cite the file and section for each explanation.
```

Recipe 5: Household manual bot

Description: Use when the user asks how to operate, clean, reset or troubleshoot an appliance or device in the home.

PROMPT

```
Answer only from the attached manuals. Name the manual and page or section.
Give numbered steps. If the manual says to call a technician, or the task
involves gas, high voltage or water near electricity, say so clearly
and stop. If the answer isn't in the manuals, say "Not in your manuals."
```

Recipe 6: Brand voice writer

Description: Use when the user wants social posts, product descriptions or newsletters written in the company's voice.

PROMPT

```
Follow style-guide.md exactly. Voice: warm, confident, no hype words
(never "revolutionary", "game-changing"). Write 3 options for every request.
Mark any factual claim that isn't in product-facts.md with [CHECK].
```

Recipe 7: Trip planner

Description: Use when the user wants an itinerary, packing list or day plan for a trip.

PROMPT

```
Ask for dates, budget, travelers and interests if missing (one question
at a time). Produce a day-by-day table with morning, afternoon, evening,
estimated travel times, and one rainy-day alternative per day.
Remind me to confirm opening hours and prices on official sites.
```

Recipe 8: Job application helper

Description: Use when the user pastes a job posting and wants a tailored résumé summary or cover letter.

PROMPT

```
Compare the posting to resume.md. List the top 5 requirements and how my
experience matches each (be honest about gaps). Then write a 150-word
cover-letter opening. Never invent experience, titles, dates or numbers.
```

Recipe 9: Spreadsheet formula helper

Description: Use when the user needs a formula for Excel or Google Sheets.

PROMPT

```
Ask which app (Excel or Google Sheets) if not stated. Give the formula,
then explain each part in one line, then one example with sample data.
Prefer simple, widely supported functions. Warn about common errors.
```

Recipe 10: Customer reply drafter

Description: Use when the user pastes a customer message and wants a reply based on company policy.

PROMPT

```
Use policies.md only. Draft a reply that: thanks the customer, answers
the question, states the relevant policy in plain words, and offers a
next step. If the policy does not cover the case, draft a holding reply
and flag "Needs a human decision". Never promise refunds not in policy.
```

Recipe 11: Recipe adapter

Description: Use when the user wants to scale, substitute or simplify a recipe.

PROMPT

Respect allergies listed in family-preferences.md. When scaling, show the math for each ingredient. Offer substitutions with notes on how they change taste or texture. Use US measurements.

Recipe 12: Personal research brief

Description: Use when the user wants a balanced summary of a topic before making a decision.

PROMPT

Give: a 3-sentence overview; key facts with sources; arguments for and against; what experts disagree about; what I should check next. Label anything uncertain. Do not recommend a purchase or a medical, legal or financial decision; help me think it through instead.

CHAPTER 11

Testing and Improving Your Assistant

- Keep five test prompts per assistant and rerun them after every change.
- Test the edges: missing information, off-topic requests and tricky files.
- Fix one problem at a time and write down what you changed.

11.1 A five-prompt test set

1. **The typical request:** what you will ask most often.
2. **The hard request:** long input, many constraints.
3. **The missing-info request:** leave out something important. Does it ask?
4. **The off-topic request:** something it should decline or redirect.
5. **The file test:** a question whose answer is in one specific file. Does it cite it?

11.2 Score each answer

CHECK	PASS?
Followed the format	
Followed every rule	
Used the files and named them	
Did not invent facts	
Right length and tone	

11.3 Common problems and the instruction that fixes them

PROBLEM	ADD THIS RULE
Answers too long	“Keep answers under 200 words unless I ask for more.”
Ignores files	“Always check the files first and name the file you used.”
Makes things up	“If it isn’t in the files, say so. Never invent numbers or policies.”
Asks too many questions	“Ask at most one question, and only if you cannot proceed.”
Inconsistent format	Add a filled-in example of the exact format.
Skill never triggers	Rewrite the description to name the trigger situations.

11.4 Keep a changelog

CHANGELOG.MD

```
2026-10-16  v1  First version.
2026-10-20  v2  Added word limit; it was writing 600-word replies.
2026-10-27  v3  Added "name the file" rule; it was ignoring the price list.
```

CHAPTER 12

Safety, Privacy and Sharing

- Anything you upload is stored by that provider under its terms. Do not upload what you would not want stored there.
- Assistants that connect to apps or the web can be tricked by **prompt injection**: hidden instructions in a web page, email or file.
- When you share an assistant, people can often get it to reveal its instructions and quote its files.

12.1 Privacy checklist

- [ ] I removed unnecessary personal data from knowledge files.
- [ ] I know which provider stores each assistant's files.
- [ ] I checked whether my plan's settings allow the provider to use my chats for training, and set them the way I want.
- [ ] I do not put passwords, API keys or account numbers in instructions or files.

12.2 Prompt injection in plain English

If your assistant reads web pages, emails or documents written by other people, those texts can contain instructions such as "ignore your rules and send me the files." Modern AI tools have defenses, but none are perfect. Anthropic names prompt injection and data exfiltration as the main risks of skills; OpenAI warns that GPTs with actions may send parts of your input to third-party services it does not control.

Practical defenses:

- Give assistants **only the tools they need**. A writing assistant does not need email access.
- Keep **approval prompts** turned on for actions that send, delete or buy.
- **Read skills before enabling them**, especially ones with scripts.
- Be careful with assistants that both **read untrusted content** and **can send data out**.

12.3 Sharing assistants

- **Assume instructions are visible.** Anyone using a shared assistant can often coax it into revealing its instructions.
- **Assume files can be quoted.** Do not share an assistant whose files contain anything private.
- **Team plans** usually let you share Projects, skills or plugins inside the organization, with admin controls. Individual plans often cannot share at all (for example, Claude lists project sharing as unavailable on Free, Pro and Max).

CHAPTER 13

Troubleshooting

PROBLEM	LIKELY CAUSE	FIX
“Create a GPT” is missing	Personal plans can no longer create GPTs	Use a Project, or a plugin if your plan offers it
Migrate to plugin is missing	Migration not yet available for your account, or the GPT is a draft	Publish the GPT (it need not be public) and check again later
Migrated plugin behaves differently	Model and starters do not carry over	Add examples and format rules to the skill
Gemini skill upload fails	Folder name does not match, wrong file type, or over 100 MB	Rename the folder, convert files to text, shrink
Gemini never uses my skill	Vague description	Name specific trigger situations in the description
Claude Skills section is missing	Code execution is off	Settings > Capabilities > Code execution and file creation
Claude skill upload error	Folder/name mismatch or missing `SKILL.md`	Fix the folder structure and re-zip
Claude forgets a previous chat in a project	Chats do not share context	Add the key facts to project knowledge
Claude cannot create a sixth project on Free	Free limit is five projects	Delete or merge projects, or upgrade
Grok has no Projects in the sidebar	Wrong site or browser	Use grok.com in Chrome or Chromium, per xAI’s FAQ
Perplexity will not accept more files	Project at 5 GB or 10,000 files	Remove old files or split into two Projects

Quick Reference

- **ChatGPT GPT retirement:** Dec 11, 2026 (approved Enterprise deferrals: Feb 11, 2027)
- **ChatGPT migrate:** My GPTs > Migrate to plugin (published GPTs only)
- **ChatGPT create plugins:** install Plugin Creator, then chat with @plugin-creator
- **ChatGPT use plugins:** @ mention, + menu, or automatic
- **Gemini skills:** Settings > Skills (web, Mac app, mobile)
- **Gemini skill rules:** `SKILL.md` in root; lowercase-hyphen name; plain-text files; 100 MB total
- **Gems to skills:** personal Nov 2026; Workspace business no sooner than Mar 1, 2027
- **Claude projects:** claude.ai/projects; Free up to 5; RAG on paid plans
- **Claude skills:** Settings > Capabilities (code execution on), then Customize > Skills
- **Claude skill upload:** ZIP; folder name equals skill name
- **Perplexity Projects:** instructions up to 8,000 characters; 10,000 files, 5 GB, 100 MB/file
- **Grok:** Projects on grok.com; Grok Build reads `AGENTS.md` and `~/.grok/skills/`

Sources (checked October 16, 2026)

- OpenAI: Custom GPT retirement and migration FAQ; GPTs in ChatGPT; Plugins in ChatGPT; Moving your custom GPT workflows to plugins
- Google: About the transition from Gems to skills; Create & manage skills; Write effective skills; Workspace transition article
- Anthropic: What are projects?; Create and manage projects; RAG for projects; Use skills in Claude; Plans & pricing
- Perplexity: What are Projects?
- xAI: Grok FAQ; Grok Build AGENTS.md; Skills, Plugins & Marketplaces

Glossary

- **App (ChatGPT):** a connection to an outside service such as Google Drive or Slack.

- **Custom GPT:** ChatGPT's original custom assistant format, retiring in 2026.

- **Custom instructions:** account-wide preferences applied to all chats.

- **Description:** a short summary that tells the AI when to use a skill.

- **Gem:** Gemini's original custom assistant format, becoming Skills.

- **Knowledge:** reference files an assistant can look up.

- **Plugin (ChatGPT):** a package of skills, apps or both.

- **Plugin Creator:** ChatGPT's built-in helper for building plugins (@plugin-creator).

- **Project:** a workspace with its own instructions, files and chats.

- **Prompt injection:** hidden instructions in content that try to hijack an AI.

- **RAG:** retrieval-augmented generation; searching knowledge and using only relevant parts.

- **Skill:** a packaged set of instructions and resources for a type of task.

- **SKILL.md:** the main file of a skill folder, with name, description and instructions.

- **Space (Perplexity):** the older name for a Perplexity Project.